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Use of Lagrangian decomposition in supply chain planning

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ABSTRACT

The integrated planning of transportation of raw material, production and distribution of products of the supply chain for a large pulp company is studied. The planning period is one year and a number of time periods are considered. The tactical decisions included in the model regard transportation of raw materials from harvest areas to pulp mills, production mix and contents at pulp mills, distribution of pulp products from mills to customers either via terminals or directly, and the selection of potential orders and their levels at customers. A mixed integer linear programming model for the supply chain problem is developed. The main solution method in this paper is a Lagrangian heuristic method based on Lagrangian decomposition. The heuristic divides the problem into two subproblems representing different physical stages in the supply chain, each including several time periods. The advantage of the proposed solution approach is that it generates feasible solutions of high quality in short time. A number of cases based on real data is analysed.

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1. Introduction

The problem addressed in this paper is the supply chain for one of the world's largest suppliers of market pulp, Södra Cell AB. The supply chain considered starts at the supply sources that is forest districts and saw mills, located in southern Sweden, passes through production units, that is pulp mills and distribution centres, that is terminals, and ends at the customers' paper mills, located mainly in central Europe. The import of logs is also a potential source of raw material. Transportation and distribution are carried out by vessels, trains, lorries and barges. In comparison to the traditional approach of minimizing costs, the goal in our model is to maximize the profit (expressed as revenues minus costs). Decisions about production mix, terminal use, which orders to select and at which level, and finally which contracts to accept are included. An overview of supply chain management in the pulp and paper industry is found in [1]. The importance of using modelling and simulation methods in the pulp industry, as well as the need for further research in the area, is pointed out in [2]. The problem of planning for the wood fibre flow is presented in [3]. The supply chain for Södra Cell AB has been studied in a number of articles. A general overview of supply chain applications can be found in [4]. In [5], the second part of the supply chain, from pulp mills to customers is considered. The problem involved different kinds of ship routes, terminal location and only one time period. An adjacent problem for coordination of production and shipment decisions can be found in [6]. In [7], the whole supply chain is considered, but only one time period and only one type of a more simple ship route are taken into account. In [8], the first part of the supply chain for Södra Cell is investigated. This supply chain starts with supplying the wood and ends at the shipment ports for the pulp mills. The main focus is detailed production planning together with transportation and storage. The planning period is three months, and daily time periods are used. A supply chain problem including product

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allocation and transportation for a forest company is presented in [9]. The supply chain starts with production at mills and ends at the customer. The overall goal is to minimize the related costs using a mixed integer programming model. Another mixed integer linear programming model is presented in [10], where the goal is to minimize the total costs for blending, loading, transportation and inventory costs. Another multi-commodity, multi-period model regarding production planning and scheduling at a refinery can be found in [11]. The solution method is based on taboo search and the results have been analysed by experienced planners. The value of coordinating production and distribution is investigated by Pirkul and Jayaraman [12]. A mixed integer programming model is formulated for the capacitated plant and warehouse supply chain management problem and a Lagrangian relaxation method is used to solve the problem [12].

In the work done earlier, there is a lack of models that take the overall supply chain into consideration, and where several time periods are used in order to consider variations in demand and supply and controlling which products that are produced at mills in different time periods. In this paper we have developed a model for this purpose for the Södra Cell problem described above. The model is detailed since it is to be used in real planning situations. The main use of the proposed model is in the annual budget process. The model becomes large and it is necessary to be able to decompose it in order to get a solution within reasonable time limits for the entire planning period. The basic problem definition is the same in this paper as in [13] but different solution approaches have been used. In [13], a Lagrangian heuristic method that relaxes all storage balance constraints leading to one subproblem per time period, is presented and tested. A weakness with the solution method was that it had, for some instances, problems to find feasible solutions when the demand changed over time. To establish a feasible solution in short time is vital in the budget process and hence we have developed a decomposition approach which better takes this aspect into account. The proposed method is based on Lagrangian decomposition, where the problem is divided into physical stages in the supply chain. This means that we consider all time periods simultaneously and changes in demand over time can be considered directly and integrated. General presentations and early papers describing Lagrangian heuristics can be found in [14–21]. We have split the flow variables from the pulp mills and then relaxed the constraints linking the old and new flow variables together. This method is called Lagrangian decomposition [22]. The method is also known as variable layering [23] and variable splitting [24]. This relaxation of constraints will divide the supply chain into two subproblems, each including all time periods. The heuristic, used in order to create feasible solutions, is a combination of a lazy heuristic [25] and a heuristic based on fixations of some of the variables in the first subproblem. By dividing the basic problem into two smaller problems, the problems get more standardized and the structure becomes simpler and it will be easier to find heuristics for solving each subproblem compared to developing a heuristic for solving the complete problem. The use of Lagrangian heuristic can be viewed as an advanced multi-start heuristic where the starting solutions are based on the solutions from the Lagrangian subproblems. In this way, we can try to generate many feasible solutions. To analyse the efficiency of the approach, we have also implemented a standard heuristic for the model.

The outline of the paper is as follows. In Section 2, the problem is described and in Section 3 some central parts of the mathematical model are presented. In Section 4, the solution methods are described. Then, in Section 5 the computational results are presented. Finally, in Section 6 some concluding remarks are made.

2. Problem

The supply chain for the Södra Cell AB is illustrated in Fig. 2.1. It starts by the acquisition and transportation of raw materials, pulp wood from forest districts and wood chips from saw mills, to the pulp mills to be used in the pulp production. A number of pulp products with different characteristics are produced according to specific recipes at five pulp mills. The final pulp products are then further distributed by shipping vessels, trains and lorries, either direct or via terminals, to the final customers, consisting of paper mills, most of them located in central Europe. The planning horizon is normally one year and it is divided into several time periods, for example, 12 months.

The wood used in the pulp production originates either from domestic sources or is imported.

In Sweden the domestic procurement is organized in forest districts, located in the southern part of the country, purchasing pulp wood from members of Södra, which is a forest association with about 35,000 members. In addition, wood chips are purchased from a number of sawmills. Imports of wood originate mainly from the Baltic States. The domestic wood is transported by lorries from the forests and saw mills to the pulp mills. Imported wood is carried by shipping vessels. The procured wood is classified into different assortments. Each assortment has unique properties which is used either individually or mixed with other assortments (according to the recipes) when producing pulp products.

Production is carried out at five pulp mills, three in Sweden and two in Norway. The amount of the different assortments needed to produce one metric ton of pulp varies. For softwood, the conversion factor is around 5 to 1 (i.e. five cubic metres of softwood is used to produce one metric ton of softwood pulp). When producing hardwood pulp less wood is needed; the conversion factor for birch is approximately 4 to 1 and for eucalyptus 3 to 1. The conversion factor wood to pulp depends also on the specific process and machinery used at the individual pulp mill. Different shares of raw materials are used to form one unit of pulp according to different recipes. All recipes cannot be used at all pulp mills because different processes and chemicals are used at the pulp mills. A production alternative describes which recipes will be used at which pulp mill and a specific cost is related to each alternative. The costs are for example consisting of change-over (or set-up) costs from using one recipe at a pulp mill to using another recipe. Only one recipe can be used at the same time at the pulp mills.

There are two alternatives to transport pulp from the mills to customers. The most important is to use shipping vessels for part of the transportation. Here, the first step is to transport pulp to the nearest harbour. Then, vessels take the pulp to

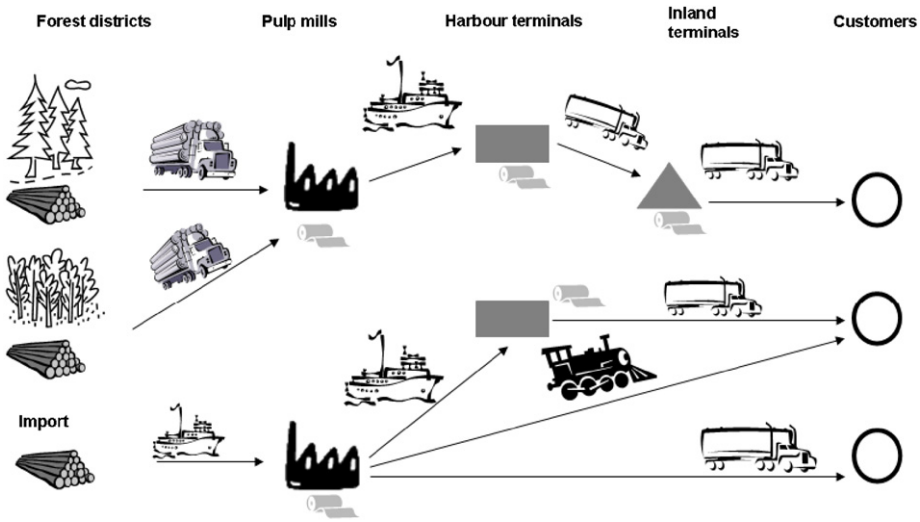


Fig. 2.1. Illustration of the supply chain for Södra Cell AB.

terminals for intermediate storage. This is then further transported by trains, lorries or barges to the final customers. The alternative is to use trains or lorries directly from mills to final customers. Södra Cell uses three shipping vessels chartered long term. They are also called TC-vessels (i.e., time chartered vessels). The routes used by the TC-vessels are divided into two groups, *A*-routes and *B*-routes. The *A*-routes, or simple routes, are routes from the relevant pulp mill to one terminal. The *B*-routes, or composite routes, are routes that start from a pulp mill, go to one terminal for unloading some of the pulp products and continue to one or more terminals for unloading the rest of the pulp products. In addition to the fleet chartered long-term, short-term vessels (i.e., spot vessels) are used. The spot vessels are chartered from one origin to one destination. Apart from these harbour terminals, there are terminals located in the interior of the continent, denoted inland terminals. They are reached by barges, trains, or lorries from harbour terminals. Each terminal has restricted capacity to receive products, depending on its size. From each terminal, the pulp products are transported to a number of customers. Terminals with storage capacities are rented on annual agreements. Trains or lorries mainly make the deliveries in Sweden and Norway. To supply the foreign customers, shipping vessels are often used, but in some cases trains and lorries can be used. There can exist agreements making these transportation modes profitable. The internal transportation from harbour terminals to inland terminals can be handled by trains, lorries or barges.

The customers are paper mills, which use the pulp products in order to make final paper products. The demand at the customers is defined in orders. Each order expresses the type of product and the related volume. A minimum and a maximum level of the volume are specified for each included product. Each order that is fulfilled will result in revenue (sales price times volume) depending on the order and the customer. A contract can include one or several orders. If a contract is accepted all the included orders have to be fulfilled. In our case, this means that a volume within the given limits (minimum and maximum) has to be delivered. In addition, there also exist free orders and fixed orders. The fixed orders have to be fulfilled independent of which contract they belong to and the free orders can be fulfilled if the sale prices are attractive enough. The delivery points are spread across Europe. In the annual budget process, the final demand is uncertain. However, many customers have been contracted over long periods and therefore the demand at customers can be estimated with relatively high accuracy. In addition, for some customers, Södra Cell operates a vendor managed inventory and then the demand is fixed in long-term contracts. Furthermore, Södra Cell is a major producer and can often affect the demand on the market by changing the prices or rebates of the pulp products.

3. Mathematical model

The mathematical model, denoted [P1], is a large mixed integer programming (MIP) model. In this section, the variables are presented and the constraints are briefly discussed. The complete mathematical model can be found in Appendix. There are two sets of variables. The binary variables represent the strategic decisions. The variables concerning production alternatives can be expressed as

$$z_a = \begin{cases} 1, & \text{if production alternative } a \text{ is used at the pulp mills,} \\ 0, & \text{otherwise.} \end{cases}$$

The variables for terminal use can be expressed as

$$z_j^T = \begin{cases} 1, & \text{if terminal } j \text{ is used,} \\ 0, & \text{otherwise.} \end{cases}$$

Finally, the binary variables for contracts at customers can be expressed as

$$z_c = \begin{cases} 1, & \text{if contract } c \text{ is accepted,} \\ 0, & \text{otherwise,} \end{cases}$$

and the binary variables for the simple routes (A-routes) can be expressed as

$$u^A_{kt} = \begin{cases} 1, & \text{if } A\text{-route } k \text{ is used in time period } t, \\ 0, & \text{otherwise.} \end{cases}$$

The second set of variables contains continuous variables representing the different flows

w_{mdiprt} = volume of raw material m transported from district d to pulp mill i for making product p according to recipe r in time period t,
y^T_{hlpt} = flow of product p from harbour terminal h to inland terminal l in time period t,
y^{tot}_{jt} = total flow of products at terminal j in time period t,
y^Q_{jsqpt} = flow of product p from terminal j to delivery point s according to order q in time period t,

and the production

u_{iprt} = production of product p at pulp mill i according to recipe r in time period t,

and finally the different kinds of variables connected to storage in the forest, at the pulp mills, at the harbour terminals and at the inland terminals, respectively

L^F_{dmt} = volume of raw material m that is stored at district d at the end of time period t,
L^M_{ipt} = volume of product p that is stored at pulp mill i at the end of time period t,
L^{HT}_{jpt} = volume of product p that is stored at harbour terminal j at the end of time period t, and
L^{LT}_{jpt} = volume of product p that is stored at inland terminal j at the end of time period t.

The flow variables on A-routes and B-routes represent the use of the long-time chartered vessels, TC-vessels, and the variables on spot flows represent the use of short-time chartered vessels. Restrictions on individual flow variables are that they have to be non-negative. The flow variables from pulp mills are presented below:

x^A_{kijpt} = flow of product p on A-route k from pulp mill i to terminal j in time period t
x^B_{kijpt} = flow of product p on B-route k from pulp mill i to terminal j in time period t
x^S_{ijpt} = spot flow of product p from pulp mill i to terminal j in time period t
y^{train}_{isqpt} = flow of product p to delivery point s according to order q from pulp mill i transported by train in time period t
y^{lorry}_{isqpt} = flow of product p to delivery point s according to order q from pulp mill i transported by lorry in time period t.

In order to model the time restriction for the routes, we introduce a variable describing the return flows on routes from harbour terminals back to pulp mills. As the shipping vessel is empty this flow does not include any products. The return flows on routes are expressed in the variables

x^R_{jit} = return flow from terminal j to pulp mill i in time period t.

The mathematical model includes several constraint types. Only one production alternative can be chosen (see Appendix constraint (A.9)). There are constraints that make sure that the supply of raw materials is not exceeded (see Appendix constraints (A.1)) and constraints that are needed in order to fulfil the demand (see Appendix constraints (A.13)–(A.16)). There are also balance constraints for forest districts, pulp mills, harbour terminals, and inland terminals, respectively (see Appendix constraints number (A.2), (A.8), (A.10), (A.20)–(A.25)). There are constraints that limit the raw materials depending on which recipe is used and capacity constraints for production, terminals, vessels, and trains, respectively, and constraints that express the limitations of storage (see Appendix constraints number (A.3)–(A.7), (A.11), (A.12), (A.17)–(A.19), (A.25)–(A.27)). The objective is to maximize the total profit. The total profit can be expressed as the total revenues minus the total costs. The sales are defined by price per order and product times the flow of products to the customers by trains and lorries. The total profit is denoted z and can be expressed as

$$\begin{aligned} z^{\text{P1}} = \text{Sales} - (&C^{\text{raw material}} + C^{\text{recipes}} + C^{\text{alternatives}} + C^{\text{trains}} + C^{\text{lorry}} + C^{\text{routes}} \\ &+ C^{\text{return}} + C^{\text{spot}} + C^{\text{flow}} + C^{\text{fixed terminal}} + C^{\text{terminal to terminal}} + C^{\text{distribution}} + C^{\text{storage}} + C^{\text{storage terminal}}). \end{aligned}$$

A more detailed description of the objective function and its included parts can be found in Appendix.

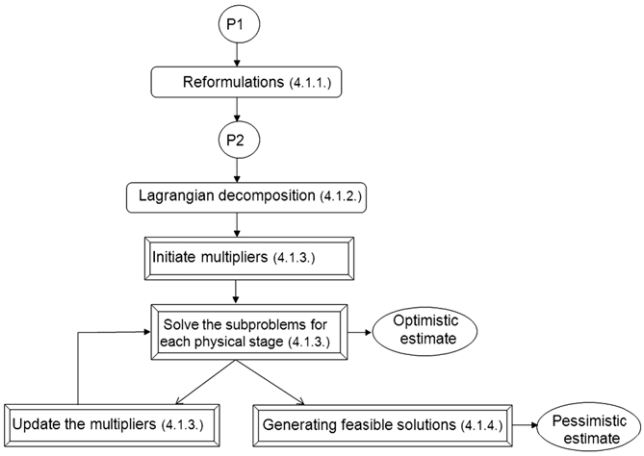


Fig. 4.1. The procedure of the Lagrangian heuristic.

4. Solution methods

A Lagrangian heuristic method and a heuristic based on division into physical stages will be described in this section.

4.1. Lagrangian heuristic method

The main solution method in this paper is a Lagrangian heuristic method. At first, we will give a general description of the procedure depicted in Fig. 4.1. Later on, in this chapter, all the squares in Fig. 4.1 will be described more in detail in the subchapters corresponding to each square in Fig. 4.1. The procedure for the method can be described as follows. First some reformulations of [P1] are made in order to be able to use the Lagrangian decomposition approach. Once the reformulations have been made to [P1], we call the model [P2]. The optimal solution of [P2] will, of course, be the same as for [P1]. The next step is to relax constraints leading to decomposition into subproblems, one for each physical stage. The Lagrangian heuristic is essentially a subgradient optimization method. This starts by initializing the Lagrangian multipliers. Then, each subproblem is solved separately. Due to the fact that we have a maximization problem, the total objective function value of the subproblems will be a higher value compared to the optimal objective value for the original problem, and can therefore be used as an optimistic estimate. A heuristic uses the solutions from the subproblems in order to produce (if possible) feasible solutions. Any feasible solution can be used as a pessimistic estimate. The subgradient method uses the solution to compute a subgradient, which is used to update the values of the Lagrange multipliers. Then the subproblems are solved using the new multiplier values and this result in new estimates. The process of updating the multipliers is iterative and will continue until the estimates are close enough to each other or after a fixed number of iterations.

The Lagrangian decomposition method described in [22,24] is used. The approach is to duplicate variables and relax the constraints holding the original and the duplicated ones at the same level. The duplication of variables will make the problem larger, but on the other hand, it enables a decomposition of the problem into appropriate parts. The main principle behind this methodology is described below.

(i) Assume that we have the problem

$$\min c^t x, \quad \text{s.t. } Ax = b,\ Bx = d,\ x \in X.$$

(ii) Duplicate the variables and add constraints with $(c = c_x + c_y)$

$$\min c_x^T x + c_y^T y, \quad \text{s.t. } Ax = b,\ By = d,\ x = y,\ x \in X,\ y \in X.$$

(iii) Relax $x = y$ with multipliers u

$$\min c_x^T x + c_y^T y + u^T(x - y), \quad \text{s.t. } Ax = b,\ By = d,\ x = y,\ x \in X,\ y \in X.$$

(iv) The problem is separable in x and y. We can use a subgradient method to find the optimal values of the multipliers u.

$$\min(c_x + u)^T x, \quad \text{s.t. } Ax = b,\ x \in X \quad \text{and} \quad \min(c_y - u)^T y, \quad \text{s.t. } By = d,\ y \in X.$$

4.1.1. Reformulations of the model

In the first part of this section new variables and constraints, added to the model, are presented. Thereafter constraints added to the model in order to improve the practical rate of convergence and get better bounds are presented. We duplicate

the flow variables from the pulp mills resulting in the following new variables:

x^{A2}_{kijpt} = flow of product p on A-route k from pulp mill i to terminal j in time period t
x^{B2}_{kijpt} = flow of product p on B-route k from pulp mill i to terminal j in time period t
x^{S2}_{ijpt} = spot flow of product p from pulp mill i to terminal j in time period t
$y^{\text{train2}}_{isqpt}$ = flow of product p to delivery point s according to order q from pulp mill i transported by train in time period t
$y^{\text{lorry2}}_{isqpt}$ = flow of product p to delivery point s according to order q from pulp mill i transported by lorry in time period t.

The following new constraints will appear in the model due to the duplications.

$$x^{A2}_{kijpt} = x^{A}_{kijpt}, \quad \forall k \in R_A,\ i \in I_k,\ j \in J_k,\ p \in P,\ t \in T \tag{1}$$

$$x^{B2}_{kijpt} = x^{B}_{kijpt}, \quad \forall k \in R_B,\ i \in I_k,\ j \in J_k,\ p \in P,\ t \in T \tag{2}$$

$$x^{S2}_{ijpt} = x^{S}_{ijpt}, \quad \forall i \in I_k,\ j \in J,\ p \in P,\ t \in T \tag{3}$$

$$y^{\text{train2}}_{isqpt} = y^{\text{train}}_{isqpt}, \quad \forall i \in I,\ s \in S,\ q \in Q_S,\ p \in P,\ t \in T \tag{4}$$

$$y^{\text{lorry2}}_{isqpt} = y^{\text{lorry}}_{isqpt}, \quad \forall i \in I,\ s \in S,\ q \in Q_S,\ p \in P,\ t \in T. \tag{5}$$

In order to improve the practical rate of convergence the following aggregated constraints are added:

$$\begin{aligned}&\sum_{k\in R_A}\sum_{j\in J_k}\sum_{t\in T} x^{A2}_{kijpt} + \sum_{k\in R_B}\sum_{j\in J_k}\sum_{t\in T} x^{B2}_{kijpt} + \sum_{j\in J_H}\sum_{t\in T} x^{S2}_{ijpt} + \sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{train2}}_{isqpt} + \sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{lorry2}}_{isqpt}\\ &= \sum_{k\in R_A}\sum_{j\in J_k}\sum_{t\in T} x^{A}_{kijpt} + \sum_{k\in R_B}\sum_{j\in J_k}\sum_{t\in T} x^{B}_{kijpt} + \sum_{j\in J_H}\sum_{t\in T} x^{S}_{ijpt} + \sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{train}}_{isqpt}\\ &\quad + \sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{lorry}}_{isqpt}, \quad \forall i \in I,\ p \in P\end{aligned} \tag{6}$$

and

$$\begin{aligned}&\sum_{k\in R_A}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{t\in T} x^{A2}_{kijpt} + \sum_{k\in R_B}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{t\in T} x^{B2}_{kijpt} + \sum_{i\in I}\sum_{j\in J_H}\sum_{t\in T} x^{S2}_{ijpt} + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{train2}}_{isqpt}\\ &\quad + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{lorry2}}_{isqpt} = \sum_{k\in R_A}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{t\in T} x^{A}_{kijpt} + \sum_{k\in R_B}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{t\in T} x^{B}_{kijpt} + \sum_{i\in I}\sum_{j\in J_H}\sum_{t\in T} x^{S}_{ijpt}\\ &\quad + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{train}}_{isqpt} + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{t\in T} y^{\text{lorry}}_{isqpt}, \quad \forall p \in P.\end{aligned} \tag{7}$$

Constraints (6) make sure that the sum of the new variables concerning A-routes, B-routes, spot trips, trains and lorries is equal to the sum of the corresponding old variables regarding each product and each pulp mill. Constraints (7) do the same thing on a more aggregated level, the sum of the given new variables has to be equal to the sum of the old variables regarding each product. By adding the above constraints more multipliers will be associated with each type of variable. The information provided by these multipliers may improve the coordination between subproblems 1 and 2 and the rate of convergence.

4.1.2. Lagrangian decomposition

The new constraints, (1)–(7) are relaxed.
The Lagrange multipliers connected to each of the relaxed constraints are presented below:

λ^{A}_{kijpt} = Lagrange multipliers for A-routes,
λ^{B}_{kijpt} = Lagrange multipliers for B-routes,
λ^{S}_{ijpt} = Lagrange multipliers for spot trips,
$\lambda^{\text{train}}_{isqpt}$ = Lagrange multipliers for flow by trains,
$\lambda^{\text{lorry}}_{isqpt}$ = Lagrange multipliers for flow by lorries,
δ^{prod}_{p} = Lagrange multipliers for products,
$\delta^{\text{mprod}}_{ip}$ = Lagrange multipliers for production at each pulp mill.

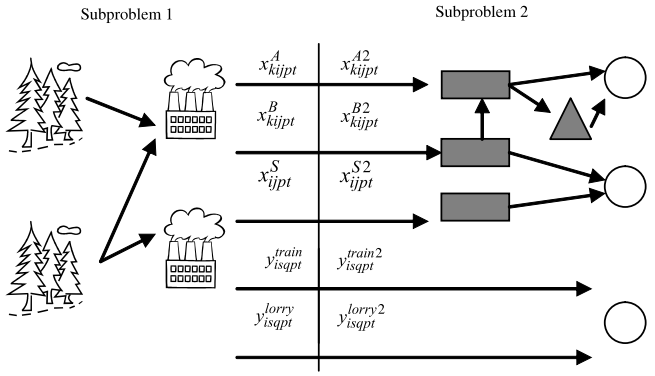


Fig. 4.2. Lagrangian decomposition resulting in two subproblems.

The value of the multipliers can be positive, negative, or zero, due to related equality constraints. The λ-multipliers are for each kind of flow for each product and the δ-multipliers relate to each product and each product at each pulp mill. The coordination of the multipliers for the separate flows and the multipliers for the aggregated constraints (6) and (7) leads to more focus on the production of each product, which is an important aspect.

The relaxation of the constraints will decompose the supply chain problem illustrated in Fig. 2.1 into two subproblems. The first problem starts at the sources (forest districts and import) and ends at the pulp mills and the second problem starts at the pulp mills and ends at the customers. The division of the supply chain problem is illustrated in Fig. 4.2.

The Lagrangian relaxed objective function can now be expressed as

$$\theta(\lambda,\delta) = \theta_1(\lambda,\delta) + \theta_2(\lambda,\delta), \quad \text{where}$$

$$\begin{aligned}
\theta_1(\lambda,\delta) \;=\; & -C^{\text{rawm}} - C^{\text{rec}} - C^{\text{alt}} - C^{\text{storFM}} - C_1^{\text{return}}/2 - \sum_{k\in R_A}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T}(c_k^A/2 + c_i^{MP}/2 - \lambda^A_{kijpt} - \delta_p^{\text{prod}} \\
& - \delta_{ip}^{\text{mprod}})x^A_{kijpt} - \sum_{k\in R_k}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T}(c_k^B/2 + c_i^{MP}/2 - \lambda^B_{kijpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})x^B_{kijpt} \\
& - \sum_{i\in I}\sum_{j\in J_H}\sum_{p\in P}\sum_{t\in T}(c_{ij}^S/2 + c_i^{MP}/2 - \lambda^S_{ijpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})x^S_{ijpt} \\
& - \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T}(c_{is}^{\text{train}}/2 - \lambda^{\text{train}}_{isqpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})y^{\text{train}}_{isqpt} \\
& - \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T}(c_{is}^{\text{lorry}}/2 - \lambda^{\text{lorry}}_{isqpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})y^{\text{lorry}}_{isqpt}
\end{aligned}$$

and

$$\begin{aligned}
\theta_2(\lambda,\delta) = {} & S - C_2^{\text{return}}/2 - C^{\text{flow}} - C^{\text{fix-term}} - C^{\text{term-term}} - C^{\text{distr}} - C^{\text{storT}} \\
& - \sum_{k\in R_A}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T}(c_k^A/2 + c_i^{MP}/2 - \lambda^A_{kijpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})x^{A2}_{kijpt} \\
& - \sum_{k\in R_k}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T}(c_k^B/2 + c_i^{MP}/2 - \lambda^B_{kijpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})x^{B2}_{kijpt} \\
& - \sum_{i\in I}\sum_{j\in J_H}\sum_{p\in P}\sum_{t\in T}(c_{ij}^S/2 + c_i^{MP}/2 - \lambda^S_{ijpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})x^{S2}_{ijpt} \\
& - \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T}(c_{is}^{\text{train}}/2 - \lambda^{\text{train}}_{isqpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})y^{\text{train2}}_{isqpt} \\
& - \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T}(c_{is}^{\text{lorry}}/2 - \lambda^{\text{lorry}}_{isqpt} - \delta_p^{\text{prod}} - \delta_{ip}^{\text{mprod}})y^{\text{lorry2}}_{isqpt}.
\end{aligned}$$

The components S, C^{rawm}, C^{rec}, C^{alt}, C^{return}, C^{flow}, $C^{\text{fix-sterm}}$, $C^{\text{term-term}}$, C^{distr}, C^{storFM}, and C^{storT} were presented in Section 3. Half of the costs for the flows from the pulp mills are charged each subproblem. The notations C_1^{return} and C_2^{return} relate to the costs for return flows in subproblem 1 and subproblem 2, respectively.

The dual problem can now be defined as $\theta^* = \min\theta(\lambda,\delta)$. For any fixed λ and δ we will get an optimistic estimate of the value of z^{P2}, which is $\theta(\lambda,\delta) \geq z^*$. We want to find the best optimistic estimate, that is to say the lowest optimistic estimate, and this problem can be solved using subgradient optimization. The subgradient optimization method is often used for non-differentiable functions.

4.1.3. Subgradient optimization

A standard subgradient method is used to find the optimal values of the Lagrangian multipliers. The first step is to choose initial Lagrangian multipliers. We have tried two different kinds of initial multipliers. The first alternative is to start with the value zero of the multipliers, later denoted ($\lambda 0$). The second alternative is to start with multiplier values set to the corresponding dual values for the relaxed constraints, in the LP solution, later denoted (λLP). The choice of step length when updating the Lagrangian multipliers follows the rule suggested by Poljak [26]. In order to avoid zig-zag behaviour, we have implemented a method that uses information of old directions when deciding new ones. This method is described in [27]. Let $d(k)$ be the direction vector used in iteration k and let $\mu(k+1)$ be the subgradient found in iteration $k+1$. By letting $d(k+1) = \mu(k+1) + \tau d(k)$ for some $0 \leq \tau \leq 1$ we get a new direction that uses information from all the previous directions. We have chosen $\tau = 0.7$, which has been shown to work well in [28].

It can be shown that the optimistic estimate of the optimal objective function yielded by a Lagrangian relaxation is always at least as good as the estimate yielded by solving the linear programming relaxation of the problem [14]. If the subproblems have the integrality property, the LP-relaxation will be at the same level as the Lagrangian relaxation of the problem. In this problem, we know that the subproblems do not have the integrality property, so we can expect a better optimistic estimate from the Lagrangian relaxation compared to the LP-relaxation estimate, although it also can produce the same value as the LP-relaxation. Since we have dualized only equality constraints we know that if the Lagrangian relaxed solution is feasible it will also be optimal.

4.1.4. Generating feasible solutions

In order to generate feasible solutions from the solutions of the Lagrangian subproblems we have used a combination of a lazy heuristic [25] and a heuristic based on fixations of variables in the first subproblem. The procedure of the heuristic is described in the following steps:

- *Step* 0. Start with the Lagrangian relaxation solution.
- *Step* 1. Check for each product if the production is greater than or equal to the sum of the minimum demand for the orders in the chosen contracts.
- *Step* 2. If the answer in *Step* 1 is yes for every product then
  - Fix the binary variables for production alternatives, terminals and contracts and solve the remaining problem.
- *Step* 3. If the answer in *Step* 1 is no regarding at least one product then
  - Fix the variables for subproblem 1 excluding the flow variables from pulp mills, and fix the binary variables for the terminals and solve the remaining problem.

4.2. Heuristic to generate solutions

We want to compare the Lagrangian heuristic with a standard heuristic. This heuristic works as follows. The supply chain is divided into two physical stages by adding a fictitious demand node. The first stage is to the pulp mills and the second stage is from the pulp mills to the customers. New variables, representing the flow from each pulp mill to the fictitious node, are added. No costs are related to this flow. The levels of the demand for each product, each time period, at the fictitious node are given by the aggregated minimum demand for the fixed orders and the aggregated maximum demand for all orders. After the problem for the first stage is solved, the new variables are fixed and each variable is forced to be equal to the total outflow from each pulp mill. Finally, the problem for the second stage is solved. There is no guarantee that the implemented heuristic will produce feasible solutions.

5. Computational results

In all implemented methods, the modelling language AMPL, version 20051214, has been used. The commercial solver CPLEX, version 10.0, with default setting, has been used in the solution process. The MIP gap expresses the accepted quality of the solution and the default setting in CPLEX is 0.01%. The instances have been solved on a 2.67 MHz Xeon processor with 2 GB RAM available.

The test cases are based on real data from Södra Cell AB. The levels and relations between the different coefficients are the same as in the real cases. Twelve time periods, each representing one month, have been used for all given cases. Three of the pulp mills are located in Sweden and one is located in Norway. Included in all cases are also ten harbour terminals and one inland terminal. There is a possibility to produce up to 19 products in total at the pulp mills. The cases 1–5 present a different number of products to produce, from only 5 products (Case 1) to 19 products (Case 5). We have also tested to include two or four different kinds of raw materials, the so-called assortments. An alternative describes which recipes will be used at which pulp mill. The recipes, in turn, describe which products will be produced and also how much of the given products that will be produced. Cases 1–5 are presented in Table 4.1. These are the same as the ones used in [13]. The separation of Case 6, into two subproblems would result in one very small subproblem and one very large. Because of this property, and the large size of this case, it is not included here. Different recipes are used at different pulp mills and according to each recipe one or two products can be produced. The *A*-routes are defined from pulp mills to terminals. Only the most common and relevant *B*-routes are included. The number of places visited is restricted to four per route, due to planning experience

Table 4.1
Description of the test cases and their sizes.

Number of	Case 1	Case 2	Case 3	Case 4	Case 5
Time periods	12	12	12	12	12
Pulp mills	4	4	4	4	4
Harbour terminals	10	10	10	10	10
Inland terminals	1	1	1	1	1
Products	5	7	10	12	19
Assortments	2	4	4	2	2
Districts	5	10	10	5	5
Recipes	9	19	34	23	33
Alternatives	2	7	15	5	5
Delivery points	10	15	30	10	10
Orders	72	72	145	84	91
Contracts	62	62	135	66	74
Routes	1 464	1 464	1 464	1 464	1 464
Continuous variables	370,811	559,115	998,052	955,284	1,545,552
Binary variables	675	680	761	682	690
Constraints	71,289	84,497	117,689	96,553	120,929

Table 4.2
Results when parts of the orders are free and the demand is not the same in each time period.

	Case 1	Case 2	Case 3	Case 4	Case 5
Optimal obj.	255,615	86,534	143,017	79,391	75,155
Time	21 min	2 h	4.2 h	288 h	6.5 h
LP-value	275,075	117,145	356,288	117,871	107,077
Time	1 min	6 min	5 min	10 min	15 min
Heuristic	inf	inf	113,976	inf	−248,698
Time	–	–	13 min	–	25 min
Lagrange, λ0, *After* 2 *itr.*					
Best dual obj.	1,129,240	1,002,700	950,206	824,449	824,412
Best primal obj.	244,301	inf	113,976	inf	−248,698
After 10 *itr.*					
Best dual obj.	1,009,076	866,977	863,098	824,449	778,35
Best primal obj.	244,301	20,023	132,144	inf	74,231
Lagrange, λLP, *After* 2 *itr.*					
Best dual obj.	273,188	113,318	247,309	113,576	105,086
Best primal obj.	inf	inf	84,084	79,391	67,894
After 10 *itr.*					
Best dual obj.	273,188	113,318	190,313	113,576	105,086
Best primal obj.	205,133	inf	143,017	79,391	67,894
Average itr. time	2 min	5 min	12 min	40 min	55 min

at the company. The demand for each time period can either be constant or vary. The total demand is the same in all test cases. Different numbers of fixed orders are also tested as well as the size of the gap between the minimum and maximum demands of the orders.

The results are presented in Tables 4.2–4.3, where the times presented are CPU-times. The average iteration time is presented for the Lagrangian heuristic. Table 4.2 presents the results when parts of the orders are free and the demand is not the same in each time period. One reason for varying demand is that the orders from customers can be with different time frequents. One customer can, for example, have six orders a year and that will lead to varying demand per month at Södra Cell AB. The properties of Cases 1–5 in Table 4.1 are the same as the properties of Cases 1d–5d in [13]. The denotation inf means that no feasible solution is found. The notation (λ0) expresses the Lagrangian heuristic started by the multipliers with the values set to zero. The notation (λLP) expresses the Lagrangian heuristic initiated with multiplier values set to the corresponding dual values for the relaxed constraints, in the LP solution. The objective function values are given in profit units, in order not to reveal the actual values.

The Lagrangian heuristic (λ0) often starts by producing bad optimistic objective values and the rate of convergence is slow. The LP-relaxation gives a better upper bound in all instances. On the other hand, the Lagrangian heuristic (λ0) often produces many good feasible solutions (pessimistic estimates). The opposite conditions apply to the Lagrangian heuristic (λLP) which in many instances produces very good upper bounds (optimistic estimates), much below the LP-value, but the feasible solutions (pessimistic estimates) are not as good. An exception is Case 4, where the Lagrangian heuristic (λLP) found the optimal solution in the first iteration. The cases in the paper come from real data and the gaps are very large. The reason

Table 4.3
Results for variants of Case 3 when the demand is the same in each time period.

	3:1, AF	3:2, AF S	3:3, PF	3:4, PF S	3:5, NF	3:6, NF S
Optimal obj.	367,156	287,943	362,019	287,456	186,571	112,056
Time	3.5 h	2.5 h	2.1 h	2.8 h	40 min	1.5 h
LP-value	597,459	494,917	545,176	457,591	364,261	253,351
Time	7 min	6 min	5 min	4 min	4 min	3 min
Heuristic	inf	267,723	311,518	269,236	153,139	94,697
Time	–	7 min	6 min	7 min	7 min	6 min
Lagrange, λ0, *After* 2 *itr.*						
Best dual obj.	1,400,310	1,092,430	1,285,060	1,094,320	972,611	867,601
Best primal obj.	inf	267,723	311,518	269,236	153,139	94,697
After 10 *itr.*						
Best dual obj.	1,155,150	1,018,210	1,120,370	1,042,740	866,541	839,706
Best primal obj.	316,655	270,723	311,518	270,236	157,312	100,274
Lagrange, λLP, *After* 2 *itr.*						
Best dual obj.	566,958	467,725	491,395	426,395	312,708	188,081
Best primal obj.	inf	220,628	140,061	213,375	101,407	inf
After 10 *itr.*						
Best dual obj.	502,366	418,743	437,743	382,163	251,831	160,646
Best primal obj.	inf	278,244	305,399	277,758	186,571	112,056
Average itr. time	9 min	8 min	14 min	11 min	12 min	13 min

for this is that the profit is optimized and this can be very different for different solutions even if the total cost is similar. The same reasoning passes for the LP gap. Even if the gaps are large, the solutions are still similar and very useful as plans for the company.

The Lagrangian heuristic (λ0) and the simple heuristic had difficulties to find feasible solutions. The Lagrangian heuristic method as well as the simple heuristic, has problem finding feasible solution in Case 2. The reason is that some of the attractive (related to high prices) free orders are impossible to fulfil with respect to the production capacity at the pulp mills. In Table 4.3, variants of Case 3 are examined. Cases 3:1 and 3:3 are the same as Cases 3a and 3b presented in [13]. The demand is the same in all time periods and the proposed method in [13] finds good solutions in these cases. We have tested different sizes of the gaps between the maximum and minimum demand levels, as well as the conditions of all orders fixed in order to test the conditions of the cases more thoroughly. The notation AF means that all orders are free, and PF means that parts of the order are free. The notation NF indicates that no orders are free. The added S, denotes when the demand gap has decreased. The results for the cases in Table 4.3 show that the proposed Lagrangian heuristic is efficient when more orders are fixed and the demand gaps get smaller in contrast to the proposed Lagrangian heuristic in [13] where the demand gaps did not have the same importance for getting good solutions. Case 3 is such that the subproblems 1 and 2 are more alike in size compared to, for example, Case 5. The proposed solution approach in [13] had problems to get good solutions in the case of very varying demand from one time period to the next time period. Since the decomposition in our new solution approach divides the supply chain into different stages, the aspect of varying demand is not as difficult to handle compared to the aspect of many free orders with very large gap between the maximum and minimum demand levels.

6. Conclusions

We have developed a large-scale model for the supply chain planning problems arising at a pulp company. The mathematical model is detailed since it is intended for direct usage. However, it is general enough to be applicable also for similar large-scale supply chain applications. We have proposed a Lagrangian heuristic method to solve the MIP model. This is based on decomposition into physical stages. The method can be used in an annual budget planning process to obtain good solutions within reasonable solution time by combining the two presented variants of the Lagrangian heuristic. A simple constructive heuristic is used for comparison.

The Lagrangian heuristic method in [13] did not generate good solutions when the demand was varying from time period to time period. The proposed Lagrangian method in this paper can generate good feasible solutions even if the demand is varying very much from time period to time period. This is an important aspect of the proposed solution method. The solutions obtained with the proposed solution method improve the solutions obtained with earlier methods.

In general, the Lagrange heuristic (λ0) provides good feasible solutions (pessimistic estimates) and the Lagrangian heuristic (λLP) provides stronger (lower) upper bounds (optimistic estimates) than the LP-relaxation. The same observations were made in [13]. The objective value from the Lagrange heuristic (λLP) can be used for evaluating the quality of the feasible solutions produced by the Lagrangian heuristic (λ0) as well as the feasible solutions produced by the simple heuristic.

The Lagrangian heuristic method produces solutions of high quality even if the demand changes during the planning period. The method is more efficient in cases where many orders are fixed and the demand gap is small. The method can be used as a complement to the method presented in [13], due to the fact that the methods have the opposite properties. This Lagrangian heuristic method produces better lower and upper bounds than that in [13] (for the instances that could be compared). This may depend on the fact that aggregated constraints are relaxed and that parts of the orders are fixed.

We include no storage of raw materials at pulp mills. That is a shortcoming because there is probably always some amount of raw materials stored at pulp mills. On the other hand, we have no restrictions for transport from forest districts to pulp mills in the model. That means that the exact amount of raw materials can be transported to the pulp mills to be used in the production process.

The model from an earlier paper made by the same authors regarding terminal location and production is annually used by Södra Cell AB in their budget process. Södra Cell AB has an intention to integrate more models concerning operations research in their production activities. This model has an ability to be one of the new implemented models for the production planning. If the model will be implemented it can be a good idea to instead of using the AMPL environment together with CPLEX which in this type of implementation is relatively slow, use direct implementation, for example in C-routines connected to the callable CPLEX system to shorten the solution time.

Appendix. The complete optimization model, [P1]

Variables

$$z_a = \begin{cases} 1, & \text{if production alternative } a \text{ is used at the pulp mills,} \\ 0, & \text{otherwise.} \end{cases}$$

$$z_c = \begin{cases} 1, & \text{if contract } c \text{ is accepted,} \\ 0, & \text{otherwise.} \end{cases}$$

$$z_j^T = \begin{cases} 1, & \text{if terminal } j \text{ is used,} \\ 0, & \text{otherwise.} \end{cases}$$

$$u_{kt}^A = \begin{cases} 1, & \text{if } A\text{-route } k \text{ is used in time period } t, \\ 0, & \text{otherwise.} \end{cases}$$

w_{mdiprt} = volume of raw material m transported from district d to pulp mill i for making product p according to recipe r in time period t

u_{iprt} = production of product p at pulp mill i according to recipe r in time period t

x_{kijpt}^A = flow of product p on A-route k from pulp mill i to terminal j in time period t

x_{kijpt}^B = flow of product p on B-route k from pulp mill i to terminal j in time period t

x_{ijpt}^S = spot flow of product p from pulp mill i to terminal j in time period t

x_{jit}^R = return flow from terminal j to pulp mill i in time period t

y_{isqpt}^{train} = flow of product p to delivery point s according to order q from pulp mill i transported by train in time period t

y_{isqpt}^{lorry} = flow of product p to delivery point s according to order q from pulp mill i transported by lorry in time period t

y_{hlpt}^T = flow of product p from harbour terminal h to inland terminal l in time period t

y_{jt}^{tot} = total flow of products at terminal j in time period t

y_{jsqpt}^Q = flow of product p from terminal j to delivery point s according to order q in time period t

L_{dmt}^F = volume of raw material m that is stored at district d at the end of time period t

L_{ipt}^M = volume of product p that is stored at pulp mill i at the end of time period t

L_{jpt}^{HT} = volume of product p that is stored at harbour terminal j at the end of time period t

L_{jpt}^{LT} = volume of product p that is stored at inland terminal j at the end of time period t.

Constants

s_{md} = supply at district d of raw material m

s_t = proportion of total supply available in time period t

a_{mi} = consumption figures of raw material m at pulp mill i to get one unit

$n_{mr}^{\max}$ = maximum proportion of raw material m in recipe r

$n_{mr}^{\min}$ = minimum proportion of raw material m in recipe r

g_i = production capacity in a 24 h period at pulp mill i

$e_{ira} = \begin{cases} 1, & \text{if pulp mill } i \text{ produces recipe } r \text{ according to alternative } a, \\ 0, & \text{otherwise} \end{cases}$

f_{pr} = proportion of product p from recipe r

k_{rp} = production a 24 h period of product p according to recipe r

m = the number of shipping vessels chartered on long-term, TC-vessels

t_k^A = unit time for A-route k

M = a large constant, in our cases, the maximum production capacity at pulp mills

N = minimum number of vessels that have to be used for flow on A-routes in a planning horizon

t_k^B = unit time for B-route k

t^{tot} = total time available for shipping vessels

t_{ji}^R = unit time for return route between terminal j and pulp mill i

s^H = shipping vessel capacity

n = share of the flow on a link in a route compared to the total flow on the given route

h_i = train transportation capacity from pulp mill i

b_j = capacity at terminal j

$d_{qpt}^{\max}$ = maximum demand according to order q for product p in time period p

$d_{qpt}^{\min}$ = minimum demand according to order q for product p in time period p

$L_{\max}^M$ = maximum storage at pulp mills

$L_{\max}^{HT}$ = maximum storage at harbour terminals

$L_{\max}^{LT}$ = maximum storage at inland terminals

v_{qp} = price of product p in order q

c_{md}^P = purchasing cost of raw material m from district d

c_{mdi}^{TR} = transportation cost of raw material m from district d to pulp mill i

c_a^{alt} = change-over costs using alternative a

c_{ir}^{rec} = cost of producing according to recipe r at pulp mill i

c_i^{MP} = unit cost for transportation from pulp mill i to the corresponding harbour

c_k^A = unit cost for A-route k

c_k^B = unit cost for B-route k

c_{ji}^R = unit cost for return route between terminal j and pulp mill i

c_{ij}^S = transportation spot cost between pulp mill i and terminal j

c_{is}^{train} = train cost between pulp mill i and delivery point s

c_{is}^{lorry} = lorry cost between pulp mill i and delivery point s

c_{hl}^T = transportation cost between terminal h and l

c_{js}^{TQ} = transportation cost between terminal j and delivery point s

c_j^T = unit cost for flow at terminal j

c_j^{FT} = fixed cost for using terminal j

c_d^{Fstor} = cost per volume unit of storing at district d

c_i^{Mstor} = cost per volume unit of storing at district d

c_j^{Tstor} = cost per volume unit of storing at terminal j.

Sets

M = The set of raw materials
D = The set of districts
I = The set of pulp mills
I_k = The set of pulp mills in route k
A = The set of alternatives
P = The set of products
R = The set of recipes
J = The set of terminals
J_H = The set of harbour terminals
J_L = The set of inland terminals
J_k = The set of terminals in route k
R_A = The set of A-routes
R_B = The set of B-routes
S = The set of delivery points
C = The set of contracts
Q = The set of orders
Q_F = The set of fixed orders
Q_C = The set of orders belonging to contract c
Q_S = The set of orders at delivery point s
T = The set of time periods

$$\begin{aligned}
\max & \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T} v_{qp} y^{\text{train}}_{isqpt} + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T} v_{qp} y^{\text{lorry}}_{isqpt} + \sum_{j\in J}\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}\sum_{t\in T} v_{qp} y^{Q}_{isqpt} \\
& - \Bigg(\sum_{m\in M}\sum_{d\in D}\sum_{i\in I}\sum_{p\in P}\sum_{r\in R}\sum_{t\in T} (c^{TR}_{mdi} + c^{P}_{md}) w_{mdiprt} + \sum_{i\in I}\sum_{p\in P}\sum_{r\in R}\sum_{t\in T} c^{\text{rec}}_{ir} u_{iprt} + \sum_{a\in A} c^{\text{alt}}_{a} z_a \\
& + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q}\sum_{p\in P}\sum_{t\in T} c^{\text{train}}_{is} y^{\text{train}}_{isqpt} + \sum_{i\in I}\sum_{s\in S}\sum_{q\in Q}\sum_{p\in P}\sum_{t\in T} c^{\text{lorry}}_{is} y^{\text{lorry}}_{isqpt} \\
& + \sum_{k\in R_A}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T} (c^{A}_{k} + c^{MP}_{i}) x^{A}_{kijpt} + \sum_{k\in R_B}\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}\sum_{t\in T} (c^{B}_{k} + c^{MP}_{i}) x^{B}_{kijpt} \\
& + \sum_{j\in J_H}\sum_{i\in I}\sum_{t\in T} c^{R}_{jt} x^{R}_{jit} + \sum_{i\in I}\sum_{j\in J_H}\sum_{p\in P}\sum_{t\in T} (c^{S}_{ij} + c^{MP}_{i}) x^{S}_{ijpt} \\
& + \sum_{j\in J}\sum_{t\in T} c^{T}_{i} y^{\text{tot}}_{jt} + \sum_{j\in J} c^{FT}_{j} z^{T}_{j} + \sum_{h\in J_H}\sum_{l\in J_L}\sum_{p\in P}\sum_{t\in T} c^{T}_{hl} y^{T}_{hlpt} + \sum_{j\in J}\sum_{s\in S}\sum_{q\in Q_s}\sum_{p\in P}\sum_{t\in T} c^{TQ}_{js} y^{Q}_{isqpt} \\
& + \sum_{d\in D}\sum_{m\in M}\sum_{t\in T} c^{\text{Fstor}}_{d} L^{F}_{dmt} + \sum_{i\in I}\sum_{p\in P}\sum_{t\in T} c^{\text{Mstor}}_{i} L^{M}_{ipt} + \sum_{j\in J_H}\sum_{p\in P}\sum_{t\in T} c^{\text{Tstor}}_{j} L^{HT}_{jpt} + \sum_{j\in J_L}\sum_{p\in P}\sum_{t\in T} c^{\text{Tstor}}_{j} L^{LT}_{jpt} \Bigg) \\
& \times \sum_{i\in I}\sum_{p\in P}\sum_{r\in R}\sum_{t\in T} w_{mdiprt} \le s_{md}, \quad \forall m \in M,\ d \in D
\end{aligned} \tag{A.1}$$

$$L^{F}_{dmt-1} + s_t s_{md} = \sum\sum\sum w_{mdiprt} + L^{F}_{dmt}, \quad \forall d \in D,\ m \in M,\ t \in T \tag{A.2}$$

$$\sum_{d\in D}\sum_{p\in P} w_{mdiprt}/a_{mi} \ge n^{\min}_{mr} \sum_{p\in p} u_{iprt}, \quad \forall m \in M,\ i \in I,\ r \in R,\ t \in T \tag{A.3}$$

$$\sum_{d\in D}\sum_{p\in P} w_{mdiprt}/a_{mi} \le n^{\max}_{mr} \sum_{p\in p} u_{iprt}, \quad \forall m \in M,\ i \in I,\ r \in R,\ t \in T \tag{A.4}$$

$$\sum_{r\in R}\left(\left(\sum_{p\in P} u_{iprt}\right) \Big/ \sum_{p\in P} k_{rp}\right) = g_i/|T| \quad \forall i \in I,\ t \in T \tag{A.5}$$

$$\left(\sum_{p\in P} u_{iprt}\right) \Big/ \sum_{p\in P} k_{rp} \le g_i/|T| \sum_{a\in A} e_{ira} z_a, \quad \forall i \in I,\ r \in R,\ t \in T \tag{A.6}$$

$$\sum_{m\in M}\sum_{d\in D}\sum_{p\in P} w_{mdiprt}/a_{mi} = \sum_{p\in P} u_{iprt}, \quad \forall i \in I,\ r \in R,\ t \in T \tag{A.7}$$

$$f_{pr}\sum_{p\in P}u_{iprt}=u_{iprt},\quad \forall i\in I,\ p\in P,\ r\in R,\ t\in T \tag{A.8}$$

$$\sum_{a\in A}z_a=1 \tag{A.9}$$

$$L^M_{ipt-1}+\sum_{r\in R}u_{iprt}=\sum_{s\in S}\sum_{q\in Q_S}y^{\text{train}}_{isqpt}+\sum_{s\in S}\sum_{q\in Q_S}y^{\text{lorry}}_{isqpt}+\sum_{k\in R_A}\sum_{j\in J_k}x^A_{kijpt}+\sum_{k\in R_B}\sum_{j\in J_k}x^B_{kijpt}+\sum_{j\in J_H}x^S_{ijpt}+L^M_{ipt},\quad \forall i\in I,\ p\in P,\ t\in T \tag{A.10}$$

$$\sum_{p\in P}L^M_{ipt}\le L^M_{\max},\quad \forall i\in I,\ t\in T \tag{A.11}$$

$$\sum_{p\in P}L^{LT}_{jpt}\le L^{LT}_{\max},\quad \forall j\in J_L,\ t\in T \tag{A.12}$$

$$\sum_{j\in J}y^Q_{jsqpt}+\sum_{i\in I}y^{\text{train}}_{isqpt}+\sum_{i\in I}y^{\text{lorry}}_{isqpt}\le z_c d^{\max}_{qpt},\quad \forall c\in C,\ s\in S,\ q\in Q_C\bigcap Q_S,\ p\in P,\ t\in T \tag{A.13}$$

$$\sum_{j\in J}y^Q_{jsqpt}+\sum_{i\in I}y^{\text{train}}_{isqpt}+\sum_{i\in I}y^{\text{lorry}}_{isqpt}\ge z_c d^{\min}_{qpt},\quad \forall c\in C,\ s\in S,\ q\in Q_C\bigcap Q_S,\ p\in P,\ t\in T \tag{A.14}$$

$$\sum_{j\in J}y^Q_{jsqpt}+\sum_{i\in I}y^{\text{train}}_{isqpt}+\sum_{i\in I}y^{\text{lorry}}_{isqpt}\le d^{\max}_{qpt},\quad \forall s\in S,\ q\in Q_F\bigcap Q_S,\ p\in P,\ t\in T \tag{A.15}$$

$$\sum_{j\in J}y^Q_{jsqpt}+\sum_{i\in I}y^{\text{train}}_{isqpt}+\sum_{i\in I}y^{\text{lorry}}_{isqpt}\le d^{\max}_{qpt},\quad \forall s\in S,\ q\in Q_F\bigcap Q_S,\ p\in P,\ t\in T \tag{A.16}$$

$$L^M_{ipt}=0,\quad \forall i\in I,\ p\in P,\ t=|T| \tag{A.17}$$

$$\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}x^A_{kijpt}\ge (Ns^H/|T|)u^A_{kt},\quad \forall k\in R_A,\ t\in T \tag{A.18}$$

$$\sum_{i\in I_k}\sum_{j\in J_k}\sum_{p\in P}x^A_{kijpt}\le Mu^A_{kt},\quad \forall k\in R_A,\ t\in T \tag{A.19}$$

$$\sum_{p\in P}L^{HT}_{jpt-1}+\sum_{k\in R_A}\sum_{i\in I_k}\sum_{p\in P}x^A_{kijpt}+\sum_{k\in R_B}\sum_{i\in I_k}\sum_{p\in P}x^B_{kijpt}+\sum_{i\in I}\sum_{p\in P}x^S_{ijpt}=y^{\text{tot}}_{jt},\quad \forall j\in J_H,\ t\in T \tag{A.20}$$

$$y^{\text{tot}}_{jt}=\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}y^Q_{jsqpt}+\sum_{h\in J_L}\sum_{p\in P}y^T_{jhpt}+\sum_{p\in P}L^{HT}_{jpt},\quad \forall j\in J_H,\ t\in T \tag{A.21}$$

$$\sum_{p\in P}L^{LT}_{jpt-1}+\sum_{h\in J_H}\sum_{p\in P}y^T_{hjpt}=y^{\text{tot}}_{jt},\quad \forall j\in J_L,\ t\in T \tag{A.22}$$

$$y^{\text{tot}}_{jt}=\sum_{s\in S}\sum_{q\in Q_S}\sum_{p\in P}y^Q_{jsqpt}+\sum_{p\in P}L^{LT}_{jpt},\quad \forall j\in J_L,\ t\in T \tag{A.23}$$

$$L^{HT}_{jpt-1}+\sum_{k\in R_A}\sum_{i\in I_k}x^A_{kijpt}+\sum_{k\in R_B}\sum_{i\in I_k}x^B_{kijpt}+\sum_{i\in I}x^S_{ijpt}=\sum_{s\in S}\sum_{q\in Q_S}y^Q_{jsqpt}+\sum_{l\in J_L}y^T_{jlpt}+L^{HT}_{jpt},\quad \forall j\in J_H,\ p\in P,\ t\in T \tag{A.24}$$

$$L^{LT}_{jpt-1}+\sum_{h\in J_H}y^T_{hjpt}=\sum_{s\in S}\sum_{q\in Q_S}y^Q_{jsqpt}+L^{LT}_{jpt},\quad \forall j\in J_L,\ p\in P,\ t\in T \tag{A.25}$$

$$L^{HT}_{jpt}=0,\quad \forall j\in J_H,\ p\in P,\ t=|T| \tag{A.26}$$

$$L^{LT}_{jpt}=0,\quad \forall j\in J_L,\ p\in P,\ t=|T|\,. \tag{A.27}$$

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